



Vasco Rocha Bartolomeu

Data Scientist · AI Engineer · Cloud & MLOps

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PROFILE

Data Scientist and AI Engineer with an M.Eng. in Data Science and a B.Eng. in Physics Engineering. I design and ship cloud-native machine-learning systems end to end — from data pipelines and model development to MLOps, infrastructure-as-code and full-stack delivery. Hands-on across the Azure and AWS stacks, with a strong track record building secure, automated platforms for clients in software, industry and sport.

SNAPSHOT

Current role	Data Scientist / AI Engineer at Enari GmbH (since 07/2023)
Education	M.Eng. Data Science & Engineering · B.Eng. Physics Engineering
Born	05.09.2000, Portugal
Languages	Portuguese (native) · English (fluent) · Spanish (fluent)

CLOUD SKILLS

Azure stack	Azure DevOps, Azure Databases, Dynamics 365, Databricks, Data Lakehouse, API Management, Azure ML / AI Studio, Application Gateway, CosmosDB, Azure Functions, Key Vault, Keycloak, ADB2C
AWS stack	EC2, S3, DynamoDB, Lambda
DevOps / CI/CD	Azure DevOps, GitLab, AWS CodePipeline, GitHub, Terraform, Terragrunt, Helm, pytest, pylint
Databases	SQL, NoSQL, MongoDB, MariaDB, MySQL, SQL Server, Postgres, CosmosDB
Containers	Docker, Kubernetes (AKS)

OTHER IT SKILLS

Languages	Python, JavaScript, TypeScript, R
Frameworks	pySpark, React, LitElement
Web frameworks	FastAPI, Flask



Data analysis	NumPy, Pandas, Matplotlib, Seaborn
Machine learning	Scikit-learn, OpenCV (Computer Vision), PyTorch, TensorFlow, A/B testing, time-series prediction & classification, CNNs, vector databases
NLP / LLMs	OpenAI API, Hugging Face, Claude AI, Llama, LlavaNextVideo, BERT, Transformer architectures, Whisper
Operating systems	Windows, Linux (Ubuntu), macOS
Scripting	PowerShell, Bash, Zsh
MS Office	Word, Excel, PowerPoint, PowerApps, Power Automate, Power BI
SEO	Google Analytics, SEMrush
CAD	AutoCAD

PROFESSIONAL EXPERIENCE

07/2023 – present Enari GmbH

AI Engineer

- Pre-processing (encoding, cleaning, normalising) with NumPy, Matplotlib, Pandas, Scikit-learn and Python.
- Data warehousing and ETL pipeline development; data processing with Azure Databricks and Azure Data Lakehouse.
- Realisation of end-to-end machine-learning pipelines and automated cloud processes via DevOps.
- Implementation and validation of ML / deep-learning architectures with PyTorch, Scikit-learn and OpenCV.
- Integration and validation of Large Language Models using OpenAI and Claude AI.
- ML / DL model deployment with Docker and development of full-stack user-data-management applications.

04/2022 – 07/2023 Earlier engineering work

Software & Systems Developer

- Built frontend applications and backend APIs connecting existing company apps to new services and databases.
- Redesigned microcontroller operating systems for integration into a home-automation system.
- Restructured computer-vision algorithms for security-surveillance systems.

SELECTED PROJECTS

05/2024 – present Enari GmbH

Azure Developer

- Developed a schedule-planning application (Docker, Azure Web App, Microsoft Dynamics 365).
- Designed a security framework for Microsoft Power Platform (Azure AD, RBAC) and built React frontends.
- Configured CI/CD pipelines (Azure DevOps); provisioned Azure ML workspaces with VNet data separation using Terraform, Terragrunt and GitLab.
- Set up Key Vaults, IAM roles, compute nodes with startup scripts and backup-enabled storage accounts in Azure.

10/2024 – 05/2024 Enari GmbH

AI Engineer

- Designed an ML algorithm for hand-pose estimation — predicting hand-joint keypoint coordinates (MediaPipe, OpenCV).
- Applied 1D/2D CNNs to electrical-impedance measurement sequences to infer hand pose from muscle activation.



- Data pre-processing and pipelines with NumPy and Pandas; ML engineering and deployment via Azure ML, versioned in Azure DevOps.

02/2024 – 10/2024 Red Bull

Azure Developer

- Built a SurfAI Coach application generating coaching messages from training-video input (Hugging Face, PyTorch, OpenCV).
- Developed an authorization model for Microsoft Dataverse with granular data- and object-level permissions (Azure Key Vault, Azure AD).
- Provisioned Azure ML workspaces with VNet isolation (Terraform, Terragrunt, GitLab) and secured Key Vaults, RBAC/IAM roles, compute nodes and backup-enabled storage.
- Implemented CI/CD pipelines for ML workflows and infrastructure (GitLab CI/CD); built React frontends.

04/2024 – 09/2024 Digital Loop

AI Engineer

- Developed LLM integrations to surface insight on SEO KPIs (Gemini, OpenAI, Mistral, Google Analytics, SEMrush).
- Infrastructure development and automated pipeline deployment with Azure DevOps, Terraform and Docker.
- Deployed Azure ML workspaces with strict VNet-based data isolation; set up secure Azure resources and CI/CD pipelines.
- Backend API handling with FastAPI; project management in Jira and Confluence.

01/2024 – 09/2024 HybridDigital

Azure Developer / Machine Learning Engineer

- Designed and deployed a Kubernetes (AKS) cluster hosting an image-preprocessing service for material classification.
- Configured Azure CI/CD pipelines for secure deployment of ERP and CRM components (Azure DevOps, Dynamics 365).
- Implemented data-handling and storage on Azure Data Lakehouse; used computer vision (OpenCV) for anomaly detection and area calculation.
- Ran A/B testing to improve detection performance; backend APIs with Flask and React frontends.

07/2023 – 01/2024 Porsche eBike

Machine Learning Engineer

- Built an ML pipeline predicting bicycle KPIs via regression models (Scikit-learn).
- Integrated A/B models to classify rider riding states and improve performance.
- Infrastructure provisioning and deployment (Azure ML, AKS).

02/2023 – 06/2023 Academic Project

Data & Machine Learning Engineer

- Correlated glass-bottle production-process data with defect counts; data ingestion via MySQL and transformation.
- Feature engineering in Python and ML models predicting the proportion of defective bottles from production parameters.

04/2023 – 06/2023 Raceland Automação

Software Developer

- Created a serverless website to minimise costs using Next.js, deployed via Docker to Google Cloud Run, domain managed through Cloudflare.

01/2023 – 03/2023 Raceland Automação

Software Developer

- Built a time-attendance manager with an asynchronous Python API handling schedules, layoff days, employee lists and hierarchy (SQL Server).
- Version control and CI/CD pipelines for automated deployment using DevOps.



04/2022 – 06/2023 **Raceland Automação**

Full-Stack Software Developer

- Engineered flexible home-automation software with Python (backend) and LitElement / Web Components (frontend).
- Used CMake to build firmware deployment for motorhome automation; shipped companion apps to the App Store and Google Play via webview.
- Integrated Siemens S7-1200 PLCs for motorhome control and combined YOLOv5 and OpenAI Whisper for object and voice recognition.

EDUCATION

09/2022 – 09/2024 **Universidade do Porto**

M.Eng. Data Science & Engineering — Faculdade de Engenharia

09/2018 – 02/2022 **Universidade do Porto**

B.Eng. Physics Engineering — Faculdade de Ciências